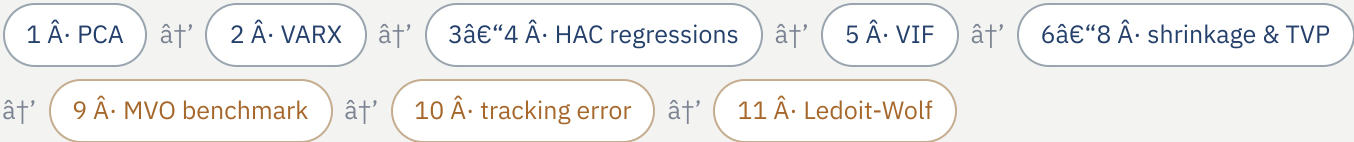


REFERENCE COMPANION

# Factor Pipeline Mathematics

The estimators, filters, and optimization problems behind every stage in the macro-factor research engine and the portfolio benchmarking pipeline – one formula block per script, in the same order they run.

Pairs with `project_architecture_map.md`  
– that file maps the scripts and data flow; this one derives the math each script implements.



## 01 Category PCA

`quant_model_macro_1.py`

Each economic category's raw series are standardized, then collapsed to a single factor via the first principal component.

STANDARDIZE EACH SERIES  $c$

$$z_{t,c} = \frac{x_{t,c} - \bar{x}_c}{s_c}$$

SAMPLE COVARIANCE OF THE STANDARDIZED BLOCK, EIGENDECOMPOSITION

$$\Sigma = \frac{1}{T-1} Z^\top Z = V \Lambda V^\top, \quad \Sigma v_k = \lambda_k v_k$$

## FACTOR SCORE AND EXPLAINED VARIANCE

$$f_t = Z_t v_1, \quad \text{EVR}_1 = \frac{\lambda_1}{\sum_k \lambda_k}$$

Run separately per category (Growth, Inflation, Rates, etc.) rather than once across all 25 series – a pooled PCA finds the single biggest common wobble across every channel and drowns out category-specific variation.

## 02 VARX(p): FF5 driven by macro factors

quant\_model\_macro\_2.py

The 5 FF factors as endogenous variables, their own lags plus the 9 macro PCA factors as exogenous regressors.

## VARX(P)

$$y_t = c + \sum_{i=1}^p A_i y_{t-i} + B x_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \Sigma_\varepsilon)$$

$y_t = [\text{Mkt-RF}, \text{SMB}, \text{HML}, \text{RMW}, \text{CMA}]^\top$ ,  $x_t =$  the 9 macro PCA factors, lag order  $p$  chosen by minimizing AIC over a max-lag search:

## LAG-ORDER SELECTION

$$\text{AIC}(p) = \ln |\hat{\Sigma}_\varepsilon(p)| + \frac{2}{T} k(p)$$

## 03 Multivariate regression, HAC standard errors

quant\_model\_macro\_3.py / \_4.py

Every ETF's excess return regressed on the factor set, with Newey–West corrected standard errors since monthly returns are typically autocorrelated.

REGRESSION

$$r_{i,t} - rf_t = \alpha_i + \sum_{j=1}^k \beta_{i,j} F_{j,t} + \epsilon_{i,t}$$

NEWBY–WEST (HAC) COVARIANCE OF  $\hat{\mathbf{B}}$ , BARTLETT KERNEL

$$\widehat{\text{Avar}}(\hat{\beta}) = (X^\top X)^{-1} \left[ \sum_{\ell=-L}^L w_\ell \Gamma_\ell \right] (X^\top X)^{-1}, \quad w_\ell = 1 - \frac{|\ell|}{L+1}$$

LAG TRUNCATION RULE OF THUMB, USED FOR L

$$L = \left\lfloor 4 \left( \frac{T}{100} \right)^{2/9} \right\rfloor$$

Step 3 uses the 5 FF factors only; step 4 repeats the same regression on all 14 (5 FF + 9 macro), and reports both OLS and HAC t-stats side by side per asset.

## 04 Variance Inflation Factor

vif\_check.py

Checks whether the 14-factor design matrix is collinear enough to be inflating the regression standard errors above.

FOR EACH REGRESSOR J, AUXILIARY REGRESSION ON THE OTHER  $K-1$ 

$$x_j = \gamma_0 + \sum_{m \neq j} \gamma_m x_m + u, \quad \text{VIF}_j = \frac{1}{1 - R_j^2}$$

All 14 factors came back between 1.2 and 2.2 on the full panel — well under the conventional 5–10 flag threshold, ruling out collinearity as the source of any weak-looking betas from Stage 03.

## 05 Normal — normal shrinkage

empirical Bayes

The continuous alternative to a hard significance cutoff: pull each noisy estimate toward a prior mean, weighted by how much precision it actually has.

LIKELIHOOD AND PRIOR

$$\hat{\beta}_i \mid \beta_i \sim N(\beta_i, se_i^2), \quad \beta_i \sim N(\mu_0, \tau^2)$$

POSTERIOR MEAN — PRECISION-WEIGHTED BLEND

$$E[\beta_i \mid \hat{\beta}_i] = w_i \hat{\beta}_i + (1 - w_i) \mu_0, \quad w_i = \frac{\tau^2}{\tau^2 + se_i^2}$$

EMPIRICAL-BAYES ESTIMATE OF  $\tau^2$ , POOLED ACROSS THE ASSET PANEL PER FACTOR

$$\hat{\tau}_j^2 = \max\left(0, \text{Var}(\hat{\beta}_{\cdot,j}) - \overline{se_{\cdot,j}^2}\right)$$

**On the actual panel:** Currency\_Factor ( $\hat{\tau} \approx 0.63$ ) and Commodities\_Factor ( $\hat{\tau} \approx 0.51$ ) carry real cross-sectional signal; Rates\_Yields, Global\_Monetary\_Policy, Credit\_Risk, US\_Monetary\_Policy, and CMA came back at  $\hat{\tau} \approx 0$  — their observed spread across ETFs is fully explained by estimation noise.

## 06 Ridge regression as a MAP estimate

Bayesian linear regression

The multi-regressor generalization of Stage 05 — a shared normal prior across the coefficient vector turns out to be exactly the ridge penalty.

POSTERIOR MODE OF  $B \sim N(0, T^2 I)$  UNDER A GAUSSIAN LIKELIHOOD

$$\text{Loss}(\beta) = \|y - X\beta\|^2 + \lambda \|\beta\|^2, \quad \lambda = \frac{\sigma^2}{\tau^2}$$

CLOSED-FORM SOLUTION

$$\hat{\beta}_{\text{MAP}} = (X^\top X + \lambda I)^{-1} X^\top y$$

With VIFs near  $10^2$  (Stage 04), the 14-factor design is close enough to orthogonal that this multivariate shrinkage behaves nearly like 14 independent Stage-05 problems, one per factor.

## 07 Hierarchical hyperprior on $\tilde{I}_j$ ,

Half-Student-t

$\tilde{I}_j$ , itself gets a prior instead of being fixed, so its uncertainty propagates into the betas  $\hat{\beta}_j$  with a heavy-tailed density that stays well-behaved as  $\tilde{I}_j \rightarrow 0$ .

HALF-STUDENT-T DENSITY,  $x > 0$ 

$$p(x) = \frac{2 \Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2}) \sqrt{\nu\pi} \text{scale}} \left[ 1 + \frac{1}{\nu} \left( \frac{x}{\text{scale}} \right)^2 \right]^{-\frac{\nu+1}{2}}$$

NON-CENTERED PARAMETERIZATION  $\hat{\beta}_j$  AVOIDS NEAL'S FUNNEL WHEN  $T_j \rightarrow 0$ 

$$\beta_{\text{raw}} \sim N(0, 1), \quad \beta = \mu + \tau \cdot \beta_{\text{raw}}$$

Chosen over an Inverse-Gamma prior on  $\tau^2$   $\hat{\beta}_j$  conjugate, but known to push  $\tau$  away from zero, which is exactly where several factors in Stage 05 actually sit.

## 08 TVP factor model $\hat{\beta}_j$ Kalman filter, FFBS, MH-within-Gibbs

tvp\_ffbs\_gibbs.py

Betas allowed to evolve over time as a random walk, estimated by Carter & Kohn (1994) forward-filtering / backward-sampling, with a Metropolis step for the one non-conjugate block.

## STATE-SPACE MODEL

$$\beta_t = \beta_{t-1} + \eta_t, \quad \eta_t \sim N(0, Q) \quad y_t = X_t^\top \beta_t + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma^2)$$

## KALMAN PREDICT / UPDATE " THE RICCATI RECURSION

$$\begin{aligned} P_{t|t-1} &= P_{t-1|t-1} + Q \\ K_t &= P_{t|t-1} X_t (X_t^\top P_{t|t-1} X_t + \sigma^2)^{-1} \\ \beta_{t|t} &= \beta_{t|t-1} + K_t (y_t - X_t^\top \beta_{t|t-1}) \\ P_{t|t} &= (I - K_t X_t^\top) P_{t|t-1} (I - K_t X_t^\top)^\top + K_t K_t^\top \sigma^2 \end{aligned}$$

CARTER-KOHN BACKWARD SAMPLING,  $T = T^{\wedge^2}, \hat{\epsilon}_t, 0$ 

$$J_t = P_{t|t} P_{t+1|t}^{-1}, \quad \beta_t | \beta_{t+1} \sim N\left(\beta_{t|t} + J_t(\beta_{t+1} - \beta_{t+1|t}), P_{t|t} - J_t P_{t+1|t} J_t^\top\right)$$

METROPOLIS-WITHIN-GIBBS FOR  $\tau_j$  " LOG-SCALE RANDOM WALK, JACOBIAN-CORRECTED

$$\log \tau_j' = \log \tau_j + \varepsilon, \quad \varepsilon \sim N(0, \text{step}^2) \quad \alpha = \min\left(1, \frac{p(\tau_j' | \cdot) \tau_j'}{p(\tau_j | \cdot) \tau_j}\right)$$

$\hat{\Gamma}^2$ ,  $\hat{\Gamma}^{1/4}$ , and  $\hat{\Gamma}^{\wedge^2}$  are all conjugate Gibbs blocks;  $\hat{\Gamma}_{,,j}$  under a Half-Student-t prior is the one block without a closed-form full conditional, hence the MH step above rather than a direct draw.

## 09 Max-Sharpe strategic benchmark

construct\_benchmark\_mvo.py

One fixed, never-rebalanced weight vector over the 5 investable FF factors, chosen to maximize Sharpe ratio subject to a long-only, market-tilt constraint.

#### OBJECTIVE AND CONSTRAINTS

$$\max_w \frac{w^\top \mu - r_f}{\sqrt{w^\top \Sigma w}} \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad 0 \leq w_i \leq 1, \quad 0.60 \leq w_{\text{Mkt-RF}} \leq 0.70$$

$\hat{\mu}$  and  $\hat{\Sigma}$  are the annualized full-sample mean and covariance of the 5 FF factor returns (monthly  $\tilde{A} - 12$ , monthly cov  $\tilde{A} - 12$ ); solved via SLSQP. Only the 5 FF factors are eligible  $\hat{\Sigma}$  they're the only long-short streams in the set that are actually replicable as a holding.

## 10 Tracking error $\hat{\Sigma}$ ex-ante and ex-post

`portfolio_factor_exposures.py`

Two independent checks on the same quantity: one from the estimated factor exposures, one from realized returns.

#### ACTIVE EXPOSURE AND EX-ANTE (ANALYTIC) TRACKING ERROR

$$\Delta\beta = \beta_{\text{portfolio}} - \beta_{\text{benchmark}}, \quad \text{TE}_{\text{ante}} = \sqrt{\Delta\beta^\top \Sigma \Delta\beta}$$

#### EX-POST (REALIZED) TRACKING ERROR

$$\text{TE}_{\text{post}} = \sqrt{12} \cdot \text{std}(r_{\text{portfolio},t} - r_{\text{benchmark},t})$$

$\hat{\Sigma}_{\text{portfolio}}$  is the market-value-weighted sum of each holding's own FF5 regression betas (full portfolio and ex-SNOW versions);  $\hat{\Sigma}_{\text{benchmark}}$  is exact by construction  $\hat{\Sigma}$  [0.60, 0, 0, 0.40, 0]  $\hat{\Sigma}$  since the benchmark *is* those weights on those factors, not an estimate.

## 11 Ledoit-Wolf shrinkage covariance

`benchmark_tracking_error_model.py`

A better-conditioned covariance estimate for the rolling tracking-error model “shrinks” the sample covariance toward a structured target.

SHRINKAGE ESTIMATOR

$$\hat{\Sigma}_{LW} = \delta^* F + (1 - \delta^*) S$$

$S$  is the sample covariance of the factor returns,  $F$  a structured target (e.g. constant-correlation), and  $\hat{\delta}^*$  the closed-form shrinkage intensity that minimizes expected quadratic loss (Ledoit & Wolf, 2004) “estimated automatically by `sklearn.covariance.LedoitWolf` rather than a hand-picked shrinkage constant.

**Scope:** currently exercised only against the synthetic placeholder portfolio in this script “swap in real weighted holdings returns before treating its tracking-error output as live.

Factor Pipeline Mathematics “companion to `project_architecture_map.md`

Python projects • `portfolio-mgmt-demo`